Spotify Hit Prediction

Group Members: Alex Wong, Aradhya Gupta, Bobby Bui, Mia Patil, Xinwei Chen

Introduction

This report analyzes the 2010s Spotify Hit Predictor dataset from Kaggle (6,398 songs, 18 features) to answer: "Can Spotify audio features predict whether a song becomes a hit?" We used six features: danceability, energy, liveness, loudness, tempo, and chorus proportion, and achieved a 79% test accuracy with a Random Forest model. We found that loudness, energy, and danceability are the primary predictors of whether a Spotify song becomes a hit.

Body

Feature engineering

We standardized *tempo* (BPM) and *loudness* (dB) using z-scores. We converted the duration from milliseconds to seconds. We engineered *chorus_proportion* (chorus start/total song duration). *Danceability*, *energy and liveness* were already scaled from 0-1. Our selected features included *danceability, energy, liveness, loudness_scaled, tempo_scaled, and chorus_proportion*. Imputation was unnecessary because there were no missing values.

Danceability and *tempo* follow fairly normal distributions, while *energy*, *liveness*, and *chorus_proportion* are skewed. Most songs have moderate to high danceability, high energy, low liveness, similar tempos, short chorus proportions, and high loudness.

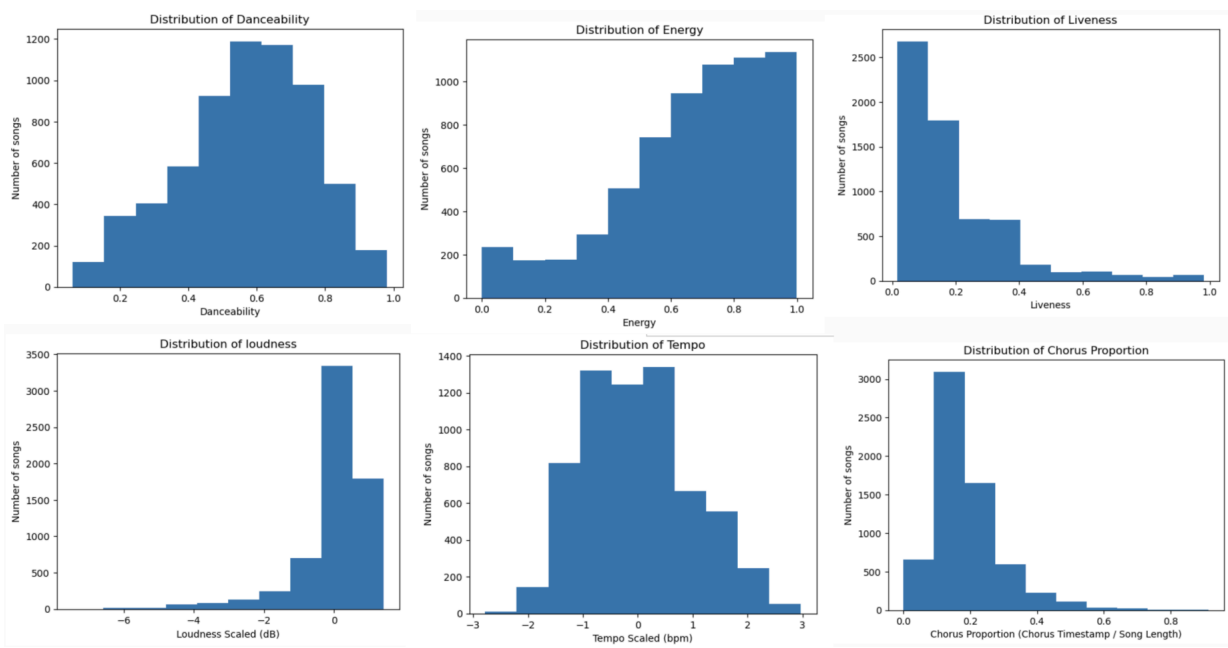

Fig 1: Distribution of Variables of Interest

We split the dataset into train/val/test (80/10/10) with stratification to preserve the target variable distribution. Our binary variable was "hit" (1) or "flop" (0) based on the author's criteria derived from Billboard chart genre considerations.

Model Selection and Hyperparameter Tuning

Among SVM, Logistic Regression with/without LASSO, KNN, Decision Tree, and Random Forest, we selected the Random Forest model because it yielded the highest validation accuracy (77.2%), with predicted hits comprising 57% of the dataset. With GridSearchCV, we tested different data splits using 5-fold cross-validation and tuned hyperparameters (max_tree_depth = 10, min_samples_to_split_node = 5, num_trees = 200) for our model. Random Forest reduced overfitting seen in the Decision Tree, handled non-linear relationships that Logistic Regression could not, and outperformed SVM and KNN in validation accuracy. Although Random Forest is computationally expensive, its performance and generalizability made it the best model, yielding a 79.1% test accuracy.

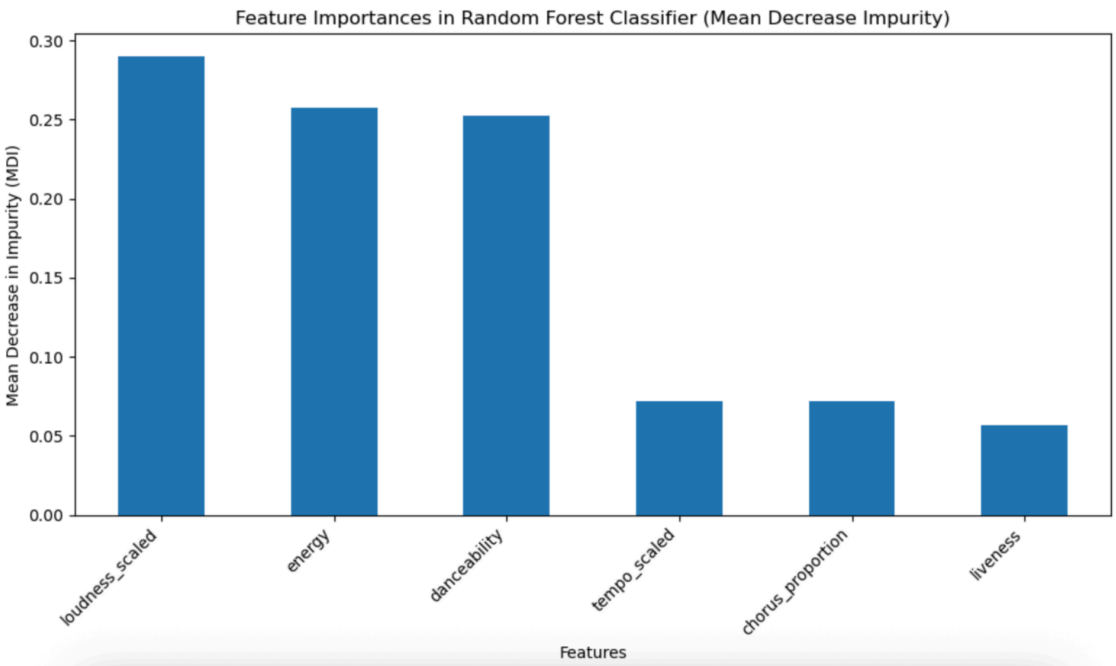

Fig 2: Feature Importances in Random Forest Classifier

We calculated each variable's mean decrease impurity to determine feature importance, finding that loudness, energy, and danceability were the strongest predictors.

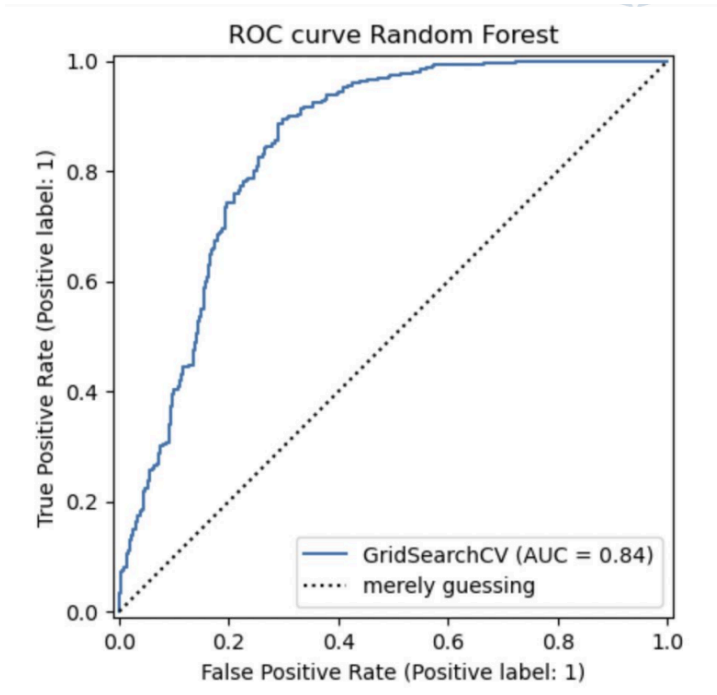

Fig 4: ROC Curve

The model's Area Under the Curve (AUC) is 0.84. This means if we randomly pick one hit and one flop, there's an ~84% chance the model assigns a higher score to the hit than to the flop. This indicates that the model is good at ordering songs from "more likely to be a hit" to "less likely". If we moved the classification threshold up or down, the Random Forest would still maintain a strong trade-off between catching real hits and not mislabeling too many flops.

```
confusion matrix:
           predict 0  predict 1
actual 0        227         93
actual 1         45        275
```

Fig 3: Confusion Matrix

According to the confusion matrix, our model correctly identifies 275 hits and 227 flops. Most errors are false positives. The important takeaway is that the model is stronger at catching actual hits (~86% correct) than at rejecting flops.

Weaknesses/Limitations

We did not include variables beyond the 6 selected or run a stepwise regression to iteratively select a subset of predictor variables. Additionally, the data from 2010–2019 may not generalize to today's music

tastes. Our dataset was also strictly limited to song metrics and lacked other potentially important predictors, such as the number of tour performances or Grammy wins.

Conclusion

Among the six tested models, Random Forest yielded the best prediction results, and our three best predictors were loudness, energy, and danceability. Our next step would be running the same statistical analysis on a more recent dataset, such as Spotify song data from the past 3 years. This helps to assess our model's generalizability to today's music.

Contributions:

Member	**Proposal**	**Coding**	**Presentation**	**Report**
Alex Wong	**1**	**1**	**0.9**	**1**
Aradhya Gupta	**1**	**1**	**1**	**0.5**
Bobby Bui	**1**	**0.8**	**0.9**	**1**
Mia Patil	**1**	**0.8**	**0.9**	**1**
Xinwei Chen	**1**	**0.8**	**0.9**	**1**